VALIDATION REPORT
OPERATIONAL QUALIFICATION
COMMUNITY SCRIPT SUITE (CORE + DOE)

JR Validated Environment — Community Script Suite

Field	Value
Document ID	JR-OQ-001
Document Version	3.0
Execution Date	2026-03-31
OQ Result	PASS — 164 / 164 tests passed, 0 failures, 0 deviations
Status	Final
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-15	Joep Rous	Initial OQ execution report. All 116 test cases passed. 2 deviations identified and closed.
2.0	2026-03-29	Joep Rous	Phase 9: 25 new numeric correctness TCs added (all PASS). DoE and core infrastructure TCs added. Total 164 test cases.
3.0	2026-03-31	Joep Rous	Regenerated from OQ evidence file. 164 test cases executed (164 passed, 0 failed). Results derived directly from evidence file.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-31
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This Validation Report documents the execution and outcome of the Operational Qualification (OQ) test suite for the community script suite (Core module and DoE module) of the JR Validated Environment. It provides objective evidence that all 164 test cases defined in JR-VP-002 v2.0 (Validation Plan) were executed, confirming that each community script produces correct, reproducible outputs for its intended use-case under 21 CFR 820.70(i).

Results are derived directly from the OQ evidence file produced by admin/admin_oq, linking each test case outcome to the raw pytest output.

2. Scope

Same as JR-VP-002 v2.0 (Validation Plan). In scope:

- 22 community R scripts and 2 community Python scripts (Core module)
- DoE scripts (jrc_doe_design.R, jrc_doe_analyse.R)
- Core infrastructure (jrc_R_hello.R, jrc_py_hello.py, integrity enforcement, bypass protection)
- Numeric correctness assertions for 9 high-risk scripts

Out of scope:

- Infrastructure scripts (bin/jrrun, admin/*) — covered by JR-IQ-001
- Non-normal distribution code paths for jrc_ss_attr and jrc_verify_attr (deferred)

3. Reference Documents

Document ID	Title	Location
JR-VP-002 v2.0	Validation Plan — OQ Community Script Suite	docs/statistics_validation_plan.pdf
JR-IQ-001	IQ Validation Evidence	docs/core_IQ_validation_plan.pdf
OQ Evidence	Community Script Suite OQ Execution Evidence (pytest output + header)	oq_execution_20260331T193715.txt

4. Test Environment

Item	Details
Execution date / time	2026-03-31T19:37:15
Host	Joeps-MacBook-Air.local
Operating system	15.7.3
R version	4.5.3
Python version	3.11.9
pytest version	pytest 8.3.4
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (dedicated OQ environment)
Test runner	admin/admin_oq
Test directory	oq/ (test_*.py files)
Test data directory	oq/data/ (synthetic datasets, fixed seeds)
Integrity hash	admin/project_integrity.sha256
OQ evidence file	oq_execution_20260331T193715.txt

5. Test Data

Synthetic datasets committed to oq/data/. No proprietary or clinical data is used. All files generated with fixed seeds for reproducibility.

File	Description	Used by
normal_n30_mean10_sd1_seed42.csv	30 values N(10,1), seed 42	jrc_normality, jrc_outliers, jrc_capability (community), jrc_descriptive, jrc_verify_attr, jrc_ss_attr, jrc_ss_sigma, jrc_ss_attr_check, jrc_ss_attr_ci
skewed_n30_lognormal_seed42.csv	30 log-normal values (skewness ≈ 0.52), seed 42	jrc_verify_attr (Box-Cox path)
outlier_n30_seed42.csv	Normal(10,1), row 15 = 5-sigma outlier	jrc_outliers
bland_altman_method1_seed42.csv	25 values N(10,2), seed 42	jrc_bland_altman
bland_altman_method2_seed42.csv	Method1 + N(0,0.2) proportional bias, seed 99	jrc_bland_altman
method1_short.csv	First 10 rows of method1 (mismatch test)	jrc_bland_altman (TC-BA-003)
weibull_n20_seed42.csv	Weibull(2,1000), 15 failures / 5 censored, seed 42	jrc_weibull
all_censored.csv	20 rows, status=0 (error path)	jrc_weibull (TC-WEIB-002)
neg_times.csv	20 rows, cycles row 1 = −50 (error path)	jrc_weibull (TC-WEIB-003)
bad_status.csv	20 rows, status includes 2 (invalid, error path)	jrc_weibull (TC-WEIB-004)
convert_multicolumn.txt	Tab-delimited, 3 headers, 20 rows	jrc_convert_txt
convert_singlecolumn.txt	200 numeric values, one per line	jrc_convert_txt
verify_attr_known.csv	30 rows: 28 × 10.0 + rows 29/30 chosen so mean=10.0, sd=1.0, K1(30,0.95,0.95)=2.220 → TL=7.780	jrc_verify_attr (TC-VER-009..011), jrc_ss_attr (TC-ATTR-008), jrc_ss_attr_check (TC-ATTRCK-004..005), jrc_ss_attr_ci (TC-ATTRCI-004)
doe_factors_*.csv / doe_results_*.csv	Multiple synthetic DoE factor and response files (full factorial, fractional, PB)	jrc_doe_design, jrc_doe_analyse (TC-DOE-DES-*, TC-DOE-ANA-*)

6. Test Execution Results

6.1 Summary

Metric	Value
Total test cases executed	164
Passed	164
Failed	0
Errors	0
OQ Result	**PASS**

6.2 Results by Test File

Test File	Scripts Covered	TCs	Result
test_ss_suite.py	jrc_ss_discrete, jrc_ss_discrete_ci, jrc_ss_attr, jrc_ss_attr_check, jrc_ss_attr_ci, jrc_ss_sigma, jrc_ss_fatigue, jrc_ss_paired, jrc_ss_equivalence, jrc_ss_gauge_rr	63	**63 / 63 PASS**
test_statistical_suite.py	jrc_bland_altman, jrc_weibull, jrc_verify_attr	22	**22 / 22 PASS**
test_diagnostic_suite.py	jrc_normality, jrc_outliers, jrc_capability (community), jrc_descriptive	17	**17 / 17 PASS**

test_gen_suite.py	jrc_gen_normal, jrc_gen_lognormal, jrc_gen_sqrt, jrc_gen_boxcox, jrc_gen_uniform	21	**21 / 21 PASS**
test_convert_suite.py	jrc_convert_csv.py, jrc_convert_txt.py	15	**15 / 15 PASS**
test_doe_suite.py	jrc_doe_design.R, jrc_doe_analyse.R	20	**20 / 20 PASS**
test_core.py	jrc_R_hello.R, jrc_py_hello.py (core infrastructure)	6	**6 / 6 PASS**
TOTAL		**164**	**164 / 164 PASS**

6.3 Numeric Correctness Test Results

The numeric correctness test cases assert computed values against independently derived expected results. Results below are read directly from the evidence file.

TC	Script	Expected value	Result
TC-DISC-006	jrc_ss_discrete	n(f=0) = 300	**PASS**
TC-DISC-007	jrc_ss_discrete	n(f=1) = 475	**PASS**
TC-DISC-008	jrc_ss_discrete	n(f=0) = 22	**PASS**
TC-DISC-009	jrc_ss_discrete	n(f=0) = 461	**PASS**
TC-DISCICI-005	jrc_ss_discrete_ci	p ≈ 0.9900 ± 0.0001	**PASS**
TC-DISCICI-006	jrc_ss_discrete_ci	p ≈ 0.8729 ± 0.0001	**PASS**
TC-ATTR-008	jrc_ss_attr	3 ≤ n ≤ 10	**PASS**
TC-ATTRCK-004	jrc_ss_attr_check	verdict = PASS	**PASS**
TC-ATTRCK-005	jrc_ss_attr_check	verdict = FAIL	**PASS**
TC-ATTRCI-004	jrc_ss_attr_ci	0.985 < p < 1.000	**PASS**
TC-SIGMA-005	jrc_ss_sigma	n = 5	**PASS**
TC-SIGMA-006	jrc_ss_sigma	n = 14	**PASS**
TC-SIGMA-007	jrc_ss_sigma	n = 3	**PASS**
TC-FAT-006	jrc_ss_fatigue	n(f=0) = 30	**PASS**
TC-FAT-007	jrc_ss_fatigue	n(f=0) = 9	**PASS**
TC-PAIRED-006	jrc_ss_paired	n = 44 (2-sided)	**PASS**
TC-PAIRED-007	jrc_ss_paired	n = 36 (1-sided)	**PASS**
TC-EQUIV-005	jrc_ss_equivalence	n = 36	**PASS**
TC-EQUIV-006	jrc_ss_equivalence	n = 45	**PASS**
TC-VER-009	jrc_verify_attr	TL ≈ 7.780 ± 0.050, verdict ✅	**PASS**
TC-VER-010	jrc_verify_attr	TL ≈ 7.780, verdict ❌	**PASS**
TC-VER-011	jrc_verify_attr	LTL ≈ 7.445, UTL ≈ 12.555, verdict ✅	**PASS**

7. Deviations from OQ Plan

All previously identified deviations (DEV-001..005) are closed. No new deviations.

ID	Affected TC(s)	Description	Status
DEV-001	TC-DISC-001	Plan used log formula (n=299); script uses chi-squared method (n=300). Justification: chi-squared is mathematically equivalent and more conservative. Assertions updated.	CLOSED
DEV-002	TC-ATTR-001..003	Plan used ±1σ spec limits; script triggers Box-Cox for skewed data causing n > cap. Tests updated to use ±3σ limits which exercise the same code path.	CLOSED
DEV-003	TC-VER-009..011	Plan had incorrect K1 derivation (1.778 vs correct 2.220). TC assertions updated before execution.	CLOSED
DEV-004	TC-PAIRED-006/007, TC-EQUIV-005/006	Plan applied ceiling() incorrectly. Correct values: 44, 36, 36, 45. Assertions updated before execution.	CLOSED

DEV-005	TC-ATTRCI-004	Plan expected proportion > 0.990 based on incorrect K1=1.778. With K1=2.220, achieved proportion = 0.9886. Assertion updated to range (0.985, 1.000).	CLOSED

8. Requirements Traceability Matrix

All URs from JR-VP-002 v2.0 mapped to executed test cases.

UR	Requirement (summary)	Test Cases	Result
UR-001	Sample size — discrete (chi-squared exact)	TC-DISC-001..009	**PASS**
UR-002	Proportion CI — exact Clopper-Pearson	TC-DISCICI-001..006	**PASS**
UR-003	Sample size — attribute tolerance interval	TC-ATTR-001..008	**PASS**
UR-004	Attribute sample size check (pass/fail boundary)	TC-ATTRCK-001..005	**PASS**
UR-005	Achieved proportion for attribute tolerance interval	TC-ATTRCI-001..004	**PASS**
UR-006	Sample size — sigma (normal approximation)	TC-SIGMA-001..007	**PASS**
UR-007	Sample size — fatigue (Weibull-adjusted)	TC-FAT-001..007	**PASS**
UR-008	Sample size — paired t-test	TC-PAIRED-001..007	**PASS**
UR-009	Sample size — equivalence (TOST)	TC-EQUIV-001..006	**PASS**
UR-010	Sample size — Gauge R&R	TC-GRR-001..004	**PASS**
UR-011	Normality testing	TC-NORM-001..005	**PASS**
UR-012	Outlier detection (Grubbs)	TC-OUT-001..004	**PASS**
UR-013	Process capability (Cp, Cpk, Pp, Ppk) — community script	TC-CAP-001..004	**PASS**
UR-014	Descriptive statistics	TC-DESC-001..004	**PASS**
UR-015	Bland-Altman method comparison	TC-BA-001..005	**PASS**
UR-016	Weibull reliability analysis	TC-WEIB-001..006	**PASS**
UR-017	Tolerance interval verification (✅/❌ verdict)	TC-VER-001..011	**PASS**
UR-018	Synthetic data generation (5 distributions)	TC-GEN-N/LN/SQ/BC/U (21 TCs)	**PASS**
UR-019	CSV file conversion (multi-column → id/value)	TC-CCSV-001..008	**PASS**
UR-020	Text file conversion (plain-text → id/value CSV)	TC-CTXT-001..007	**PASS**
UR-021	Input validation and usage messages (all scripts)	Error-path TCs across all suites	**PASS**
UR-022	RENV_PATHS_ROOT enforcement (R bypass protection)	TC-CORE-OQ-004	**PASS**
UR-023	DoE — design generation and analysis	TC-DOE-DES-001..012, TC-DOE-ANA-001..008	**PASS**
UR-024	Core infrastructure (hello scripts, integrity, bypass)	TC-CORE-IQ-001, TC-CORE-OQ-001..005	**PASS**
Total		**All 164 community OQ tests**	**PASS**

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Community Script Suite (Core + DoE modules) is complete. All 164 test cases were executed on 2026-03-31 and passed with no failures, no errors, and no deviations.

All 24 user requirements (UR-001 through UR-024) are covered by at least one passing test case. The OQ acceptance criteria stated in JR-VP-002 Section 10 are fully met.

The Community Script Suite (Core and DoE modules) is hereby declared OPERATIONALLY QUALIFIED and released for use in design verification activities in accordance with the JR Validated Environment quality system.

OQ Evidence file:

/Users/joeprous/.jrscript/MyProject/validation/oq_execution_20260331T193715.txt